

Arjun Sen

Junior developer focused on testing and debugging

SYNTHETIC DEMO MATERIAL | All people, projects and results are fictional.

Profile

Early-career developer practicing Python services and integration tests. Comfortable reproducing bugs and discussing evidence with teammates; still learning advanced PostgreSQL concurrency.

Skills

Python, Flask, SQL, PostgreSQL basics, pytest, Git, HTTP and structured logging.

Help desk ticket service

Built fictional endpoints to open, assign and close tickets. Added field validation, role checks and tests for invalid state transitions.

Reproduced a bug where retrying an HTTP request created duplicate tickets. Added request identifiers to logs, wrote a failing integration test and introduced a unique client request ID.

The integration suite contains 32 synthetic cases covering validation, permissions and retries. Load testing and database lock behavior have not yet been evaluated.

CSV import exercise

Built a preview step that reports invalid rows before saving imported tickets. Used a transaction so an invalid row cannot leave a partially saved batch.

Compared skipping bad rows with rejecting the entire batch and documented why the all-or-nothing option was chosen for the exercise.

Collaboration

Worked with a fictional designer to clarify confusing error messages. Shared a minimal reproduction and screenshots with another developer before agreeing on the fix.

Learning background

Fictional programming coursework and personal projects. Has not operated an on-call service or designed a distributed system.